

unsuccessful attempt. Figure 4 shows the output obtained after automatic evaluation of the program with test cases by CodeRunner. From the figure, it is clear that we have provided four test cases and the program has passed all four of those test cases. This is how CodeRunner can be used to set up automated programming tests. Remember, this article only provides a light introduction to CodeRunner, but for more details and advanced features, you can refer to the Moodle plugins directory.

Installing Virtual Programming Lab (VPL)

Now that we are familiar with the use of CodeRunner, it is time to introduce Virtual Programming Lab (VPL), which is much more powerful than CodeRunner. However, if both plugins can evaluate programming assignments by running the code provided by a student, what is the benefit of the additional features provided by VPL? Let us go through two academic scenarios to understand the need for the stronger features provided by VPL.

First, let us consider a student enrolled in an online course covering a new programming language, say, Rust. He has probably paid a fee and is willing to spend a lot of time to learn a new programming language. Usually, such online courses have programming assignments to test the knowledge level of the student. The student can write a program and test it to understand whether he has learned that topic or not. But is he going to copy the answer from some other source, say, the Internet? Most probably not. His aim is to learn a new programming language and plagiarising the program will not help him in any manner. In this scenario, the features of CodeRunner are sufficient and VPL might be overkill.

Now consider the second scenario in which you have to conduct an online exam, the marks for which will be considered for the final grade of a student. The academic community strongly disapproves of the practice of plagiarism but, in this scenario, some students might be tempted to get solutions online. In this case, the plugin should have additional features to check whether students have copied their programs from some other source. In this case, CodeRunner is not the best solution whereas VPL provides an option to check for plagiarism. VPL offers many such features that we should discuss. VPL is an activity module that can be used to create and conduct programming assignments in online programming courses.

	Test	Expected	Got	
✓	print f ("%d", sqr (-3))	9	9	✓
✓	print f ("%d", sqr (-11))	121	121	✓
✓	print f ("%d", sqr (5))	25	25	✓
✓	print f ("%d", sqr (6))	36	36	✓

Passed all tests! ✓

Figure 4: Automatic evaluation by CodeRunner

As mentioned earlier, the installation of VPL is a two-step process. First, we have to install the plugin and then a sandbox server. The execution server that provides a sandbox for VPL is called a jail server, which is installed in your local machine. First log in to your Moodle server as the administrator, and you can install the plugin directly from the Moodle plugins directory. To do this, select the menu item *Dashboard>Site administration>Plugins>Install plugins*. The VPL plugin is also installed from a zip file provided by the Moodle plugins directory. The installation file for the jail server can be downloaded from <http://vpl.dis.ulpgc.es/index.php/home/download>. The latest version available for download is VPL Jail Execution System 2.2.2 and the name of this installation file is *vpl-jail-system-2.2.2.tar.gz*. Extract this file in your system and open a terminal inside the directory *vpl-jail-system-2.2.2*. In this directory, there is an installation shell script named *install-vpl-sh*. The installation can be completed by running this shell script, by executing the command *./install-vpl-sh* on the terminal.

The detailed installation screenshots are given in the document *Configure.pdf*. If jail server is operating properly, it must respond with a page showing the text OK, if the URL [http://server_name\[:port\]/OK](http://server_name[:port]/OK) is typed in the address bar of a Web browser.

Using Virtual Programming Lab

Let's use VPL to set up an online programming test. Here again, we have to see how a teacher can set up a test using VPL and how a student can attend that test. To set up a test, you should log in as either an admin user, course designer or teacher, depending on the privileges provided by the admin user. In this example, we are again logging in as the admin user to simplify things. The course *Programming* created in Moodle has a topic called Topic1. The option *Add an activity or resource* allows us to add new VPL activity to Topic1. Figure 5 shows how VPL can be added as an activity.

Now, details regarding the program to be answered have to be provided — the program's name, its description, etc. Just like for CodeRunner, for VPL too, we need to set up a test with a single question, which in this case is, "Find the square of a number entered through the keyboard."

The other advantage of VPL over CodeRunner is that in the case of the latter, the solution program should be written in a specific programming language preset by the teacher. VPL does not have this restriction. Depending on the extension of the program, the required compiler will be selected by the jail server to do the compilation. For example, a file with an extension *.py* will be treated as a Python program and another file with an extension *.c* will be treated as a C program automatically by the jail server. The programming languages supported by VPL include Ada, C, C++, Fortran, Java, Pascal, Prolog, SQL, Scheme, etc.

After this, test cases should be provided to conduct automatic code evaluation. The more the test cases, the